

The section overhead provides facilities that are required to support and maintain the transportation of customer traffic safely across the network. RSOH occupies 27 octets, containing information about the frame structure required by the terminal equipment. MSOH occupies 45 octets, containing information about error correction and automatic protection switching (APS) messages (such as alarms and maintenance messages) as may be required within the network. AU pointer points to the location of the J1 byte in the payload—the first byte in the virtual container.

Reduced mux structure

The heterogeneous nature of modern network structures has made it necessary that all type of signals like PDH and ATM signals are transported over the SDH network. Some of the telecom hierarchies adopted around the different parts of the world are tabulated (see table on previous page). The process of matching the signals to the network is called mapping, and these tributary sig-



Fig. 2: Illustration for container



Fig. 3: Formation of virtual container



Fig. 4: Formation of tributary/administrative unit

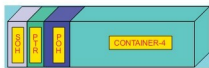


Fig. 5: Formation of STM-1 from C-4

nals are mapped with an SDH system in a defined fashion. The container is the basic package unit for tributary channels. The STM-1 frame structure allocates $260 \times 9 = 2340$ octets for a container with signal carrying capacity of $260 \times 9 \times 8 \times 8000$ (149.76Mbps). This carrying capacity is sufficient for transporting a 139.264Mbps tributary

signal (E4—European digital hierarchy). Practically, a special container (C-n) is provided for each PDH tributary signal (Fig. 2). These containers are always much larger than the payload to be transported, and the remaining capacity is used partly for justification (stuffing) in order to equalise out-timing inaccuracies in the PDH signals. Where synchronous tributaries are mapped, fixed fill bytes are inserted instead of justification bytes.

A virtual container (VC-n) is made up from the container, thus formed together with the path overhead (POH) (Fig. 3). The virtual container carried in an STM-1 frame is referred to as a virtual container level four, or VC-4, and is transported unchanged over a path through the network. The channel capacity of VC-4 is $261 \times 9 \times 8 \times 8000 = 150.34$ Mbps. The VC-4 signal's carrying capacity can also be subdivided to permit the transport of multiple lower-level PDH signals.

The next step towards the formation of a complete STM-N signal is the



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